

1D CNN — Architecture Variant Comparison

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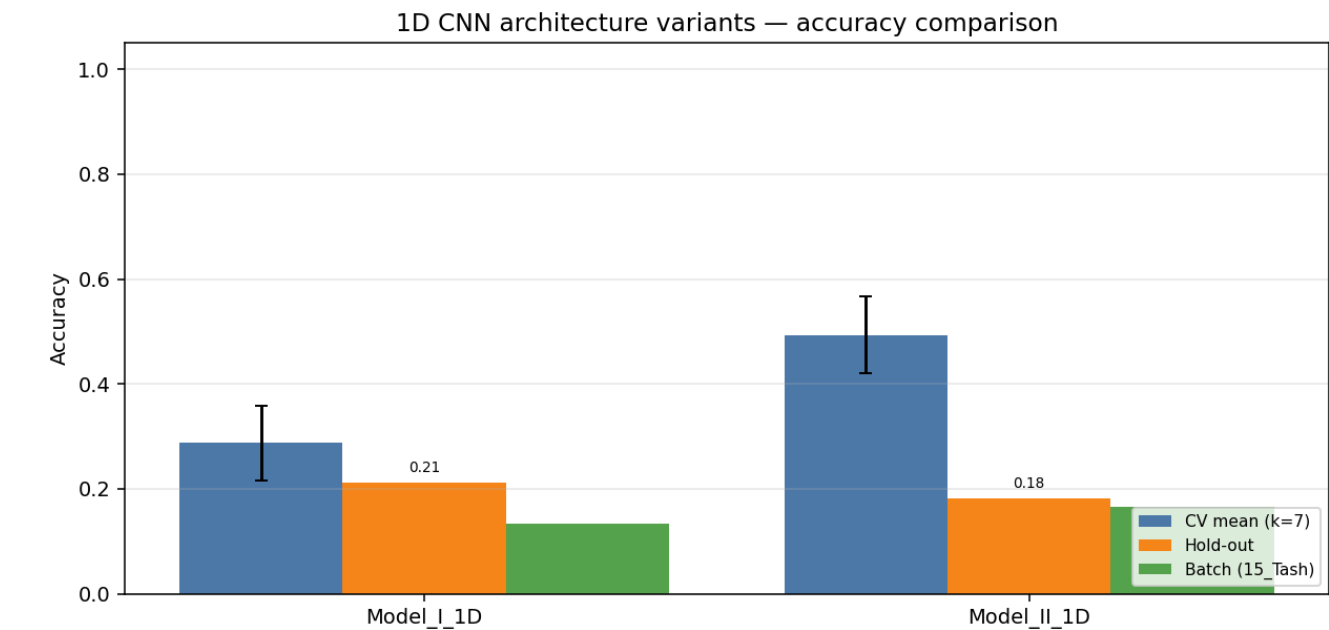
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	20 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

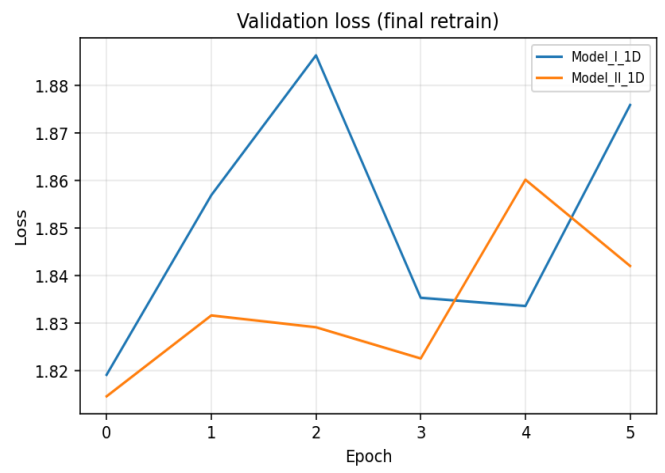
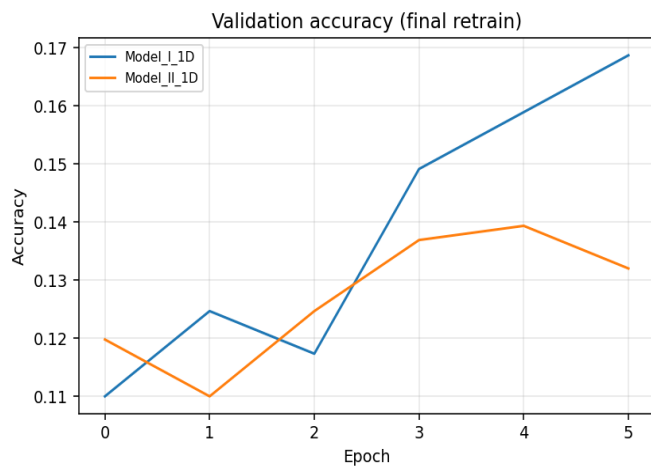
Variant comparison (summary)

Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	118,630	0.2880	0.0713	0.2121	1.7740	0.1333	4/30
Model_II_1D	73,926	0.4930	0.0735	0.1818	1.7793	0.1667	5/30

Best on hold-out: **Model_I_1D** · Best on CV: **Model_II_1D**



Training curves (final retrain on full training pool)



Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	118,630
CV mean ± std	0.2880 ± 0.0713
Hold-out test acc / loss	0.2121 · 1.7740
Batch-test acc	0.1333 (4/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.2619	0.2857	0.2619
2	1750	42	0.1667	0.2381	0.2143
3	1750	42	0.3333	0.4286	0.3333
4	1750	42	0.2381	0.3095	0.2619
5	1750	42	0.3571	0.3571	0.2857
6	1757	41	0.3902	0.3902	0.2927
7	1757	41	0.2683	0.3902	0.3902

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.000	0.000	0.000	6
Double_Raise	0.250	0.333	0.286	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.208	0.833	0.333	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	0.076	0.194	0.103	33
weighted avg	0.083	0.212	0.113	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	4	5	0.800
Drum_Roll	0	5	0.000
Overall	4	30	0.133

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM):
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	73,926
CV mean ± std	0.4930 ± 0.0735
Hold-out test acc / loss	0.1818 · 1.7793
Batch-test acc	0.1667 (5/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1750	42	0.4762	0.5238	0.4524
2	1750	42	0.5952	0.5952	0.4762
3	1750	42	0.5714	0.5714	0.5238
4	1750	42	0.4762	0.4762	0.4762
5	1750	42	0.3810	0.4048	0.4048
6	1757	41	0.5366	0.5854	0.5610
7	1757	41	0.4146	0.5366	0.3902

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.000	0.000	0.000	5
Double_Pistol_Recoil	0.182	1.000	0.308	6
Double_Raise	0.000	0.000	0.000	6
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	0.000	0.000	0.000	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	0.030	0.167	0.051	33
weighted avg	0.033	0.182	0.056	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	5	30	0.167